SUMMARY · MK 5 DECISION MAKING & NEGOTIATION · WEEK 6

# Decision Trees

Decision trees visualise sequential decisions under uncertainty: decision and chance nodes, probabilities, payoffs. Build and roll back with expected monetary value, multi-stage decisions, EVPI, risk attitude and utility, and probability sensitivity.

## Material

### 1 Elements and construction

Decision nodes (□), chance nodes (○, probabilities sum to 1), payoffs. Draw left to right in time; roll back right to left: EMV = Σ p × payoff at chance nodes, choose the best branch at decision nodes.

### 2 The example tree, redrawn Deep dive

| Decision | Chance | Payoff | EMV |
|---|---|---|---|
| Big | High 0.6 | 10 | 4.4 ✓ |
| | Low 0.4 | −4 | |
| Small ‖ | High 0.6 | 5 | 3.4 |
| | Low 0.4 | 1 | |
| None ‖ | — | 0 | 0 |

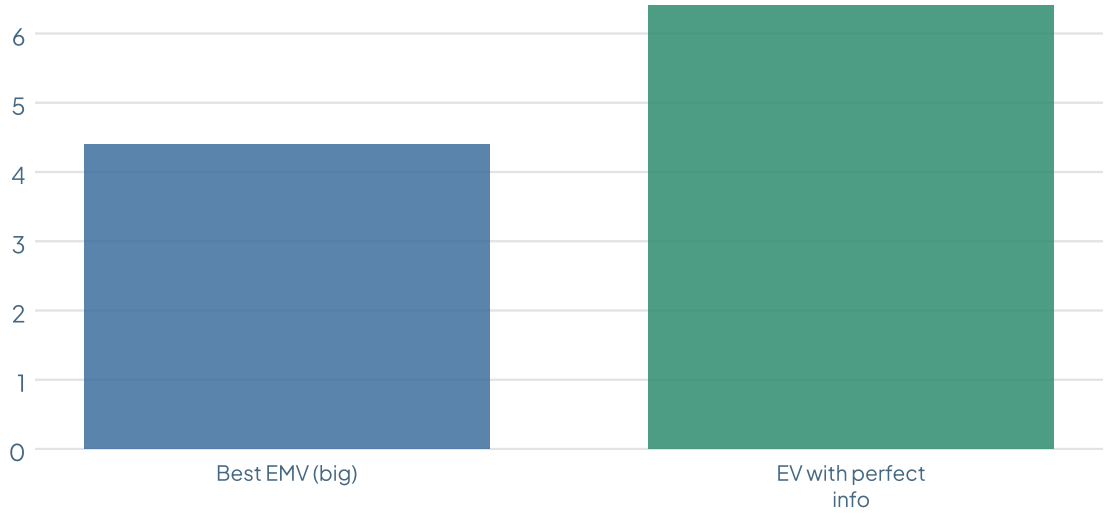
## 3 Worked example

Example: launch big EMV 4.4, small 3.4, none 0 → big. EVPI = 6.4 – 4.4 = 2.0. Indifference probability p = 0.5. Staged decisions keep valuable options open. The class example (power design, 6.75 vs 6.2) follows the same logic.

## 4 Simulator: EMV, indifference probability and EVPI Deep dive

The lines cross at p = 0.5.

**Expected value: without vs with perfect information**



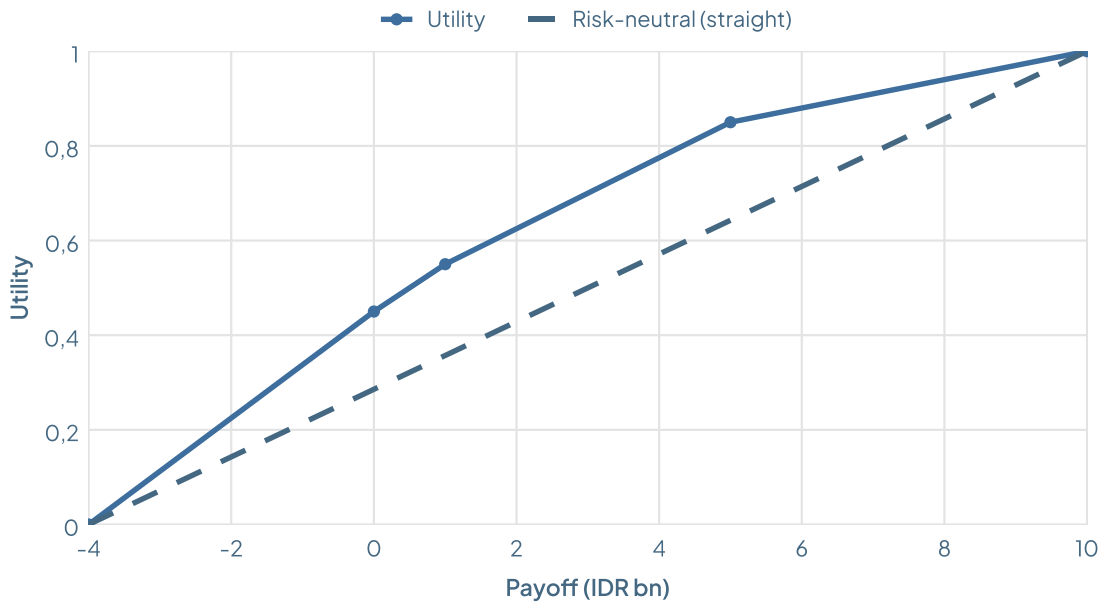

EVPI = 2.0: the ceiling on research spend.

## 5 Risk, utility and limitations

EMV assumes risk neutrality; risk-averse decision makers use utilities (probability-equivalence) and expected utility, and examine risk profiles. Trees grow bushy, probabilities are bias-prone, and one-off decisions differ from long-run averages — always test sensitivity.

## 6 Utility: why people reject the highest EMV Deep dive

**Risk-averse utility function (illustrative)**




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EU big = 0.60 · EU small = 0.73
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With a concave (risk-averse) utility, small wins despite a lower EMV.

## Key takeaways

1. Decision, chance, payoff nodes.
2. Roll back: EMV = Σ p × payoff.
3. EVPI.
4. Indifference probabilities.
5. Expected utility for risk aversion.

## Case studies

Case & worked answers

### Linking to Pelindo (MK4) and BUMDes

Use the tools to quantify your Pelindo tree (probabilities for demand and subsidence, NPV payoffs, EMV, EVPI of a geotechnical study). SMART fits many criteria with little uncertainty; trees fit uncertain events — they can be combined.

## Worked examples

Problem 1 · Medium **EMV, EVPI, dan titik impas probabilitas**

Investasi produk baru: sukses p = 0,3 → +Rp50 M; gagal → –Rp10 M. Tidak investasi = 0. (a) Keputusan EMV? (b) Berapa maksimal membayar riset sempurna? (c) Di p berapa investasi mulai layak?

1. (a) EMV investasi = 0,3 × 50 + 0,7 × (–10) = 15 – 7 = **8** > 0 → investasi.
2. (b) Dengan info sempurna: jika sukses, investasi (50); jika gagal, tidak (0). EV = 0,3 × 50 + 0,7 × 0 = 15.
3. EVPI = 15 – 8 = **7**.
4. (c) Impas ketika 50p – 10(1 – p) = 0 → 60p = 10 → p = 0,167.

**Answer:** Investasi (EMV Rp8 M); riset maksimal Rp7 M; layak selama p > 16,7%.

# Practice questions

Try answering first. The answer key is below each question.

**1. At a chance node you compute…**

A. Max payoff
B. Σ p × payoff
C. Min payoff
D. Simple mean

**Answer: B.** EMV.

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**2. EVPI is 2.0. Research costing 2.5 is…**

A. Worth it
B. Not worth it
C. Always worth it
D. Can't tell

**Answer: B.** Imperfect research is worth even less.

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**3. Rejecting the top EMV for fear of big loss is…**

A. Risk-seeking
B. Risk-neutral
C. Risk-averse
D. Irrational

**Answer: C.** Concave utility.

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**4. Big and small are equal when p(high) = …**

A. 0.4
B. 0.5
C. 0.6
D. 0.7

**Answer: B.** $14p - 4 = 4p + 1$.

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**5. Trees are drawn left to right following…**

A. Payoff size
B. Time order
C. Alphabet
D. Probability

**Answer: B.** Solved in reverse.

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## Sources

- [Complete] 06 Decision Tree (MK5) — slide berupa gambar; contoh di sini disusun ulang
- Goodwin & Wright, Bab 5–6